river faced a more difficult task. But the Chinese flagship was lost within minutes of the start of the action. The French had equipped two small steam launches with spar torpedoes (mines hung on long poles projecting from their bows). One of these exploded against *Yang Wu*'s hull, sinking the vessel. The flatiron gunboats were also sunk by the fire of French guns. Some Chinese ships ran aground attempting to flee up river. The sailing junks proved perfectly useless. In an hour every Chinese ship in Foochow had been sunk or disabled and the arsenal and docks were ablaze.

News of this display of imperial naval power was greeted with joy by a jingoistic French public. Vietnam was secured for France after some harder fighting on land. Courbet died the following year, ensuring his elevation to the status of a French national hero. France's first dreadnought, lanched in 1913, would bear his name.

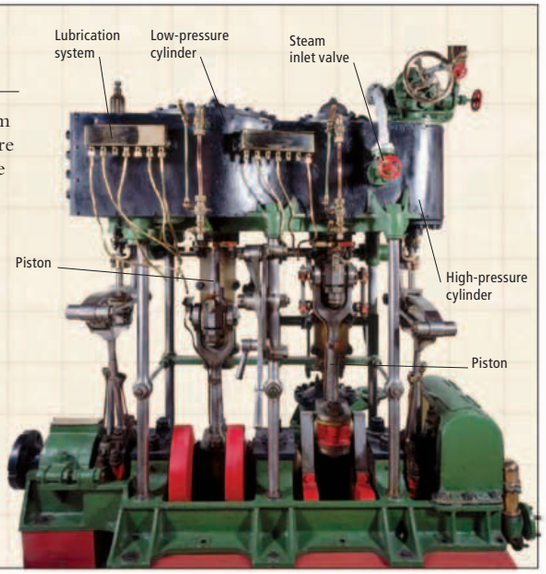
Unequal contest
French armored cruisers rain destruction down upon the wooden Chinese vessels at Foochow. All of the Chinese ships were destroyed, while the French lost none.



WEAPONS AND TECHNOLOGY

COMPOUND ENGINE

Early steam engines had a single cylinder. A simple compound steam engine is fitted with two cylinders, the first using the high-pressure steam from the boiler and the second exploiting the lower pressure steam emitted from the first cylinder as exhaust. Invention of the first truly practical two-cylinder compound steam engine is attributed to Scottish engineer William McNaught in 1845, but compound engines did not begin to be used at sea until the 1860s. "Compounding" greatly improved the efficiency of maritime engines. By increasing the distance ships could travel without refueling, compound engines made steam vessels effective for oceanic travel. By 1880 they were almost universal in new-built ships. Over the following decade triple-expansion engines (with three cylinders) began to come into use. During the 19th century innovations in steam-engine design were usually pioneered by merchant vessels, naval development following in their wake.



Engine with two cylinders
The second cylinder of a compound engine uses steam that escapes from the first. Each cylinder drives a piston.

THIRD ANGLO-BURMESE WAR

IRRAWADDY RIVER



Date November 16–28 ,1885
Forces British: c.50 river boats
Losses British: none; Burmese: unknown

Location Irrawaddy River, Burma

Britain fought three wars against the kingdom of Burma in the 19th century, the first in 1823–26, the second in 1852, and the third in 1885. The first two wars ended in compromise, but the third was followed by the annexation of Burma. The 1885 war was rooted in colonial rivalry between Britain and France. The British government suspected that Burmese King Thibaw Min was allowing his country to slip under French influence. To forestall the French they launched a military expedition to take over the country.

Major General Harry Prendergast was sent from Madras to the Burmese border with an army and naval force numbering some 9,000 men. They could not advance on land through the Burmese jungle, but the Irrawaddy River offered a clear highway from the Bay of Bengal to the royal capital of Mandalay. The British-owned Irrawaddy Flotilla Company provided river steam boats and pilots for the expeditionary force. Prendergast had brought with him 24 Maxim guns, the first ever appearance of these belt-fed machine guns in military service. They were mounted on steamers to provide formidable firepower.

King Thibaw was slow to organize his defenses. In principle the river was defended by shore batteries and forts. At Minhla, boats were to be sunk in the navigable channel as blockships, forcing the British to halt under the guns of a fort on the river bank. In practice there was little resistance. The blockships were never sunk and the shore positions were overrun by British-Indian troops and naval brigades with the support of devastating fire from the armed steamers. Wherever the guns of the steamships were trained upon the shore, the Burmese swiftly surrendered, their defeat completed by the fall of Mandalay on November 28.

Snider-Enfield rifle, 1860
This breech-loading precussion-cap rifle was adopted by the British forces in 1866 and was used during the Third Anglo-Burmese War.



ANGLO-ZANZIBAR WAR

ZANZIBAR



Date August 27, 1896
Forces British: 5 ships; Zanzibar rebels: 1 armed yacht
Losses British: none; Zanzibar rebels: 1 yacht sunk

Location Zanzibar, off east coast of Africa

The Anglo-Zanzibar conflict of 1896 was the shortest war in recorded history and the ultimate exercise of "gunboat diplomacy." The island of Zanzibar was a British protectorate, its sultan Hamad bin Thuwaini accepting the advice of a British resident. There was, however, tension over Zanzibar's persistence with slave trading, which the British wished to suppress. The political crisis that led to war came on August 25, 1896, when the sultan died. The succession to the sultanate of Britain's preferred candidate, Hamad bin Muhammed, was preempted by Hamid bin Thuwaini's nephew Khalid bin Bargash, who established himself in the royal palace with several thousand armed followers in what amounted to a coup d'état. The Royal Navy had two ships at Zanzibar, the third-class cruiser *Philomel* and the gunboat *Thrush*. These were joined by the sloop *Sparrow* and a body of Royal Marines was put ashore. Rear Admiral Harry Rawson then arrived on board the Cape Station flagship *St George*, a thoroughly modern first-class cruiser with 9.2in main armament, accompanied by the torpedo cruiser *Racoon*. The Zanzibar navy consisted of a single British-built armed yacht, *Glasgow*, which was anchored in front of the palace.

Rawson presented an ultimatum: if Bargash didn't surrender by 9:00 a.m. on August 27, the Royal Navy would open fire. The bombardment began at 9.02 a.m. The *Glasgow* was soon sunk and the royal palace and harem were pulverized. After some 40 minutes of destruction by the naval guns, Bargash fled to seek refuge at the German consulate and his followers raised the white flag. Around 500 people lay dead amid the ruins of the royal buildings. Hamud bin Muhammed became sultan and, out of gratitude to the British, abolished the slave trade. Zanzibar was also obliged to pay the cost of the war.